

$$\text{resolvability} = \log p(y|I, q) - \log p(y|\tilde{I}, q)$$

threshold

high

0

0

threshold

high

$$\text{NEED} = \log p(y|I, q) - \log p(y|\emptyset, q)$$

*needs image,
but not high-frequency*

(rare; treat as edge)

text-answerable

(skip)

READABLE EDGE

concentrate rollouts here

(the only region where RL gains capability)

UNREADABLE

withhold rollouts;
supervise abstention here

y : gold answer. $\tilde{I}$: image with fine detail destroyed. $\emptyset$: blank. anchored during training by verifiable correctness.